Scientists Remake a Cup That Changes Colour

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About 1,600 years ago, Roman artists made a glass cup that changes colour. Now scientists have finally learned how to make it again. The Lycurgus Cup sits in the British Museum in London. It is one of only 40 glass cage cups that still exist. In normal light, the cup looks **jade** green. But if you shine a light through it, it glows **ruby** red. The answer is tiny bits of gold and silver mixed into the glass. These bits are thousands of times smaller than a grain of sand. They are called **nanoparticles**. Lisa Hollands, a scientist at the University of Sheffield, worked to crack the **recipe**. At last she found a mix that works. Her glass looks green in normal light and glows red when light shines through. Glass artist Elliot Walker used the new glass to make a **sculpture** for the university's Festival of the Mind on November 6. The science may also help make new materials that control light, and special glass for storing dangerous **waste** safely.

Word Treasure

nanoparticles -- Nanoparticles are pieces of matter far too small to see.
sculpture -- A sculpture is a work of art made by shaping material.
jade -- Jade is a green stone.
ruby -- Ruby is a deep red gemstone.
recipe -- A recipe is a set of steps for making something.
waste -- Waste is stuff people throw away.

Background you need

Glass can be made in many colours.

Tiny bits of metal in glass can change the way light moves through it.

Break it down

WHO: Lisa Hollands, Caroline Jackson and Elliot Walker
WHAT: Scientists remake a 1,600-year-old colour-changing cup
WHERE: The University of Sheffield and the British Museum
WHY: To learn how the Romans made the glass

Why it matters

Old things can teach us new science, and you can see the magic cup change colour in a museum.

Test what you remember

1. How old is the Lycurgus Cup?

(A) About 600 years old
(B) About 160 years old
(C) About 16,000 years old
(D) About 1,600 years old

2. What colours does the cup show?

(A) Blue in normal light and gold when light shines through
(B) Red in normal light and green when light shines through
(C) Green in normal light and red when light shines through
(D) White in normal light and black when light shines through

3. What tiny bits of metal are in the glass?

(A) Iron and copper
(B) Tin and lead
(C) Gold and silver
(D) Zinc and nickel

4. Who worked out the recipe?

(A) Christina Metcalfe
(B) Lisa Hollands
(C) Caroline Jackson
(D) Elliot Walker

5. What did Elliot Walker make with the new glass?

(A) A telescope
(B) A mirror
(C) A window
(D) A sculpture

6. When will the sculpture be shown?

(A) At the Festival of the Mind on November 6
(B) At the British Museum on November 6
(C) At the Festival of the Mind on December 6
(D) At the Turner Museum on November 6

Answer key (try the quiz first!): 1: D 2: C 3: C 4: B 5: D 6: A

Questions to think about

MY THOUGHTS (at least 20 words):